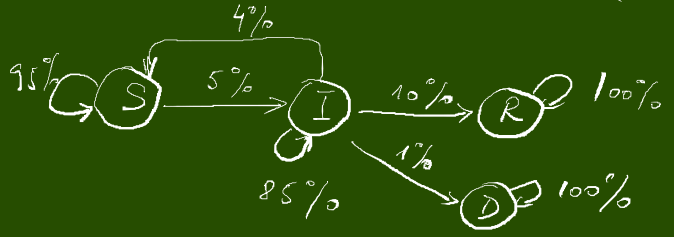
1) Sisteme dinamice liniare (TS)

SIR



$$A = \begin{array}{c} \\ S \\ I \\ R \\ D \end{array} \begin{pmatrix} S & I & R & D \\ 0{,}95 & 0{,}04 & 0 & 0 \\ 0{,}05 & 0{,}85 & 0 & 0 \\ 0 & 0{,}10 & 1 & 0 \\ 0 & 0{,}01 & 0 & 1 \end{pmatrix}$$

diagonalizabilă

$n = 0 \qquad v(0) = \begin{pmatrix} S(0) \\ I(0) \\ R(0) \\ D(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \longrightarrow v(1) = A v(0)$

$v(2) = A v(1) = A^2 v(0)$

. . . . . .

$v(n) = A^n v(0)$

$\lim_{n\to\infty} v(n) = \lim_{n\to\infty} A^n v(0)$

$A^n = ?$ dc. A diagonalizabilă $\Rightarrow A = P \cdot D \cdot P^{-1}$

$A^n = \underbrace{P D P^{-1} \cdot P D P^{-1} \cdot \ldots \cdot P D P^{-1}}_{n \text{ ori}} \cdot P \cdot D \cdot P^{-1} = P \cdot D^n \cdot P^{-1}$

$D^n = \begin{pmatrix} \lambda_1^n & & \\ & \ddots & \\ & & \lambda_m^n \end{pmatrix}$

$$A = \begin{array}{c} \\ S \\ I \\ R \\ D \end{array} \begin{array}{c} \begin{array}{cccc} S & I & R & D \end{array} \\ \begin{pmatrix} 0{,}75 & 0{,}04 & 0 & 0 \\ 0{,}07 & 0{,}85 & 0 & 0 \\ 0 & 0{,}10 & 1 & 0 \\ 0 & 0{,}01 & 0 & 1 \end{pmatrix} \end{array}$$

3) $A = \begin{pmatrix} 3 & 1 \\ -5 & -3 \end{pmatrix}$

$\lambda_1 = 2, \quad \lambda_2 = -2$

$tr(A) = 0$

$\det(A) = -4 \neq 0$

$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$

$\lambda_1 = 0 \quad \lambda_2 = 3$

$tr(A) = 3$

$\det(A) = 0$

$P_A(x) = x^2 - 3x + 0$

$= x^2 - 3x$

$\begin{vmatrix} 1-x & 1 \\ 2 & 2-x \end{vmatrix} =$

$= (1-x)(2-x) - 2 = x^2 - 3x$

1) $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $\leftarrow i$ $\leftarrow -i$

$\mathbb{R}$ $\quad$ $\mathbb{C}$

2) $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$\lambda = 0 \quad a(\lambda) = 2$

$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$P_A(x) = x^2 - 0 \cdot x + 0$

$= x^2$

$A = P \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} P^T = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ Fals!

$P_A(x) = x^2 - 2x + 1$

$= (x-1)^2$

$\lambda = 1 \quad a(\lambda) = 2$

$A = P \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} P^T = I_2$ Fals.

$$A = (a_{ij}) \quad \text{matrice stocastică}$$

$$a_{ij} \geq 0$$

$$\sum_i a_{ij} = 1$$

lanț Markov $\longrightarrow$ HMM

hidden Markov models

$$A = \begin{array}{c} \\ S \\ I \\ R \\ D \end{array} \begin{array}{c} \begin{array}{cccc} S & I & R & D \end{array} \\ \begin{pmatrix} 0{,}95 & 0{,}04 & 0 & 0 \\ 0{,}05 & 0{,}85 & 0 & 0 \\ 0 & 0{,}10 & 1 & 0 \\ 0 & 0{,}01 & 0 & 1 \end{pmatrix} \end{array}$$

$\lambda = 1 \longrightarrow a(\lambda) = 2$

$V_{\lambda_1} = Sp\{e_3, e_4\} \subseteq \mathbb{R}^4$

$g(\lambda_1) = 2$

$|\lambda_2|, |\lambda_3| < 1$

$$A^n = P \cdot \begin{pmatrix} 1 & & & \\ & \lambda_2^n & & \\ & & \lambda_3^n & \\ & & & 0 \end{pmatrix} P^{-1} \to \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 10/11 & 10/11 & 1 & 0 \\ 1/11 & 1/11 & 0 & 1 \end{pmatrix}$$

$$v(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A^n v(0) \xrightarrow{n \to \infty} \begin{pmatrix} 0 \\ 0 \\ 10/11 \\ 1/11 \end{pmatrix}$$

1) A matrice stoc $\Rightarrow \lambda = 1$ val. proprie

stare staționară $v = \begin{pmatrix} \vdots \end{pmatrix}$ $\quad Av = v$

v vect. propriu pt $\lambda = 1$



$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$v = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} \quad Av = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

$$A^n = \begin{cases} A, & n \text{ impar} \\ I_2, & n \text{ par} \end{cases} \qquad \lim_{n \to \infty} A^n v(0) \nexists$$

PERRON - FROBENIUS

3) $A=(a_{ij})_{i,j}$ $\exists n$, $A^n=(a_{ij}^{(n)})_{i,j}$ are toate componentele >0

$\Rightarrow$

a) $\lambda=1$ val. pr. pt. A cu mult. algebrică 1

b) $\forall\, v(0)\in\mathbb{R}^n$ $\lim_{n\to\infty} A^n v(0)=v$ $\exists$, indep. de $v(0)$

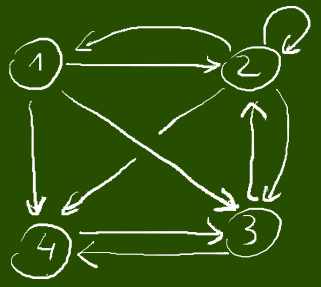
vect. pr. pt A

c) toate val. pr. ale mat. A verifică $|\lambda|<1$ pt. $\lambda=1$

$d=0{,}85$ $\forall\,\lambda\in V(A)$

$$v(1)=\left(d\cdot A\cdot v(0)+(1-d)\frac{1}{4}v(0)\right)$$

PageRank (GOOGLE)



A^3

$$A=\begin{pmatrix} 0 & 1/4 & 0 & 0\\ 1/3 & 1/4 & 1/2 & 0\\ 1/3 & 1/4 & 0 & 1\\ 1/3 & 1/4 & 1/2 & 0\end{pmatrix} \qquad v(0)=\begin{pmatrix}1/4\\1/4\\1/4\\1/4\end{pmatrix}$$

$$v(1)=Av(0)=\begin{pmatrix}1/16\\13/48\\17/48\\13/48\end{pmatrix}\rightarrow \qquad v(n)=A^n v(0)$$

$Av = v$ $\quad v = \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}$

$$\begin{cases} 1/4\, y = x \\ 1/3\, x + 1/4\, y + 1/2\, z = y \\ 1/3\, x + 1/4\, y + t = z \\ 1/3\, x + 1/4\, y + 1/2\, z = t \end{cases}$$

$\Rightarrow \quad x = \frac{1}{4} y, \quad y = t, \quad t - \frac{1}{2} z + t = z \quad 2t = \frac{3}{2} z \quad t = \frac{3}{4} z$

$$\Rightarrow \begin{cases} x = \frac{3z}{16} \\ y = \frac{3z}{4} \\ z = z \\ t = \frac{3z}{4} \end{cases}$$

$$V_1 = Sp\left\{ \begin{pmatrix} 3 \\ 12 \\ 16 \\ 12 \end{pmatrix} \right\} \Rightarrow v = \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \frac{1}{43} \begin{pmatrix} 3 \\ 12 \\ 16 \\ 12 \end{pmatrix}$$

$x + y + z + t = 1$

$\Rightarrow \quad \lim_{n \to \infty} A^n v(0) = \frac{1}{43} \begin{pmatrix} 3 \\ 12 \\ 16 \\ 12 \end{pmatrix}$ → pagina 3

$$v(1) = \underbrace{\left(dA + (1-d)\, \frac{1}{4} \begin{pmatrix} 1 & \cdots & 1 \\ 1 & \cdots & 1 \\ 1 & \cdots & 1 \\ 1 & \cdots & 1 \end{pmatrix} \right)}_{\tilde{A}} v(0)$$

d damping factor

$u \cdot v$ $\qquad$ $\langle u, v \rangle$

quantum

$k = \mathbb{R}$ $\qquad$ ($k = \mathbb{C}$)

$V \times V \longrightarrow \mathbb{C}$

$(u, v) \longrightarrow \langle u, v \rangle$

$\langle u, v \rangle = \overline{\langle v, u \rangle}$

$\mathbb{R}^n$ $\qquad$ 1) $\langle u, v \rangle = u^t v$

$\langle u, v \rangle = \langle v, u \rangle$

$u^t v \;?\; v^t u$

$(1,1)\,(1,1)$

$(u^t v)^t = v^t u$ ✓

2) $\langle u_1 + u_2, v \rangle = \langle u_1, v \rangle + \langle u_2, v \rangle$

$(u_1 + u_2)^t v = (u_1^t + u_2^t) v = u_1^t v + u_2^t v$

$= \langle u_1, v \rangle + \langle u_2, v \rangle$

3) $\langle \lambda u, v \rangle = (\lambda u)^t v = \lambda u^t v = \lambda \langle u, v \rangle$

4) $\langle u, u \rangle = u^t u = x_1^2 + \dots + x_n^2 \geq 0$

$$u = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

5) $\langle u, u \rangle = 0 \iff x_1 = \dots = x_n = 0$

$\iff u = 0$

$\mathbb{R}_n[x] \simeq \mathbb{R}^{n+1}$

exemplu. $V = \{ f : [a,b] \to \mathbb{R} \mid f \text{ cont} \}$

$$\langle f, g \rangle = \int_a^b f(t) \cdot g(t)\, dt$$

$$\langle f, f \rangle = \int_a^b \underbrace{f^2(t)}_{\geq 0} dt \geq 0$$

$f(t) = 0 \quad \forall t \in [a,b] \Rightarrow \langle f, f \rangle = 0$

$\langle f, f \rangle = \int_a^b f^2(t)\,dt = 0 \quad \xrightarrow{?} \quad f(t) \equiv 0 \ \forall t$

$f^2(t) \geq 0$

$\int_0^1 f^2(t)\,dt = 0$

$f \neq 0$

$[0,1]$

$$f(t) = \begin{cases} 0, & t \in [0, \frac{1}{2}) \\ 1, & t = \frac{1}{2} \\ 0, & t \in (\frac{1}{2}, 1] \end{cases}$$

pp. $\exists\, t_0 \in (a,b)$ a.t. $f(t_0) \neq 0$

pp $f(t_0) > 0$

f contínua $\Rightarrow \exists\, \varepsilon > 0$ a.t. $f(t) > 0$

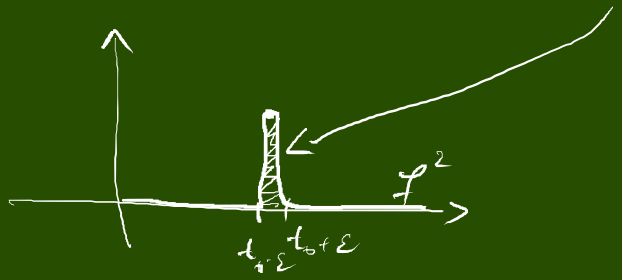
$t \in (t_0 - \varepsilon, t_0 + \varepsilon) \subseteq (a,b)$

$$0 = \int_a^b f^2(t)\,dt = \int_a^{t_0-\varepsilon} f^2(t)\,dt + \int_{t_0-\varepsilon}^{t_0+\varepsilon} f^2(t)\,dt + \int_{t_0+\varepsilon}^b f^2(t)\,dt$$

≥ 0 ; > 0 ; ≥ 0

≥ 0

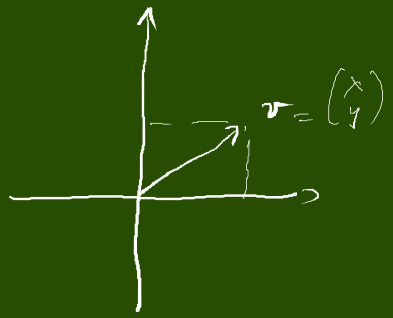
Falso



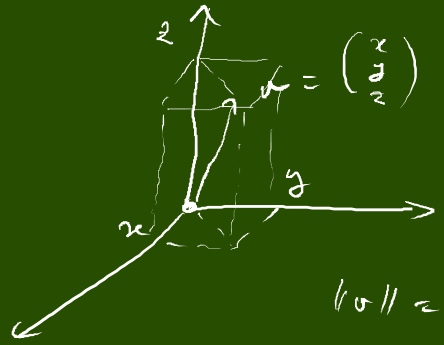
$(V, \langle , \rangle)$ $\qquad \langle u, v \rangle \geq 0$

$\|v\| = \sqrt{\langle v, v \rangle} \geq 0$

$\mathbb{R}^n$ $\qquad \|v\| = \sqrt{\sum_{i=1}^{n} x_i^2}$



$\|v\| = \sqrt{x^2+y^2}$



$\|v\| = \sqrt{x^2+y^2+z^2}$

$\mathbb{R}^n \qquad u = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad v = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$$(x_1y_1 + \dots + x_ny_n)^2 \leq (x_1^2 + \dots + x_n^2)(y_1^2 + \dots + y_n^2)$$

$\langle u, u \rangle \geq 0 \qquad \forall\, u \in V$

$\langle u + \lambda v, u + \lambda v \rangle \geq 0 \qquad \forall\, \lambda \in \mathbb{R}$

$$\langle u + \lambda u, u + \lambda v \rangle = \langle u, u + \lambda v \rangle + \lambda \langle v, u + \lambda v \rangle$$
$$= \langle u + \lambda v, u \rangle + \lambda \langle u + \lambda v, v \rangle$$
$$= \langle u, u \rangle + \underline{\lambda \langle v, u \rangle} + \underline{\lambda \langle u, v \rangle} + \lambda^2 \langle v, v \rangle$$
$$= \underbrace{\lambda^2 \|v\|^2 + 2\lambda \langle u, v \rangle + \|u\|^2}_{\geq 0 \qquad \forall\, \lambda \in \mathbb{R}}$$

$\Delta \leq 0$

$$\Delta = 4\langle u, v\rangle^2 - 4\|u\|^2 \cdot \|v\|^2 \leq 0$$

$$\Rightarrow \quad \langle u, v\rangle^2 \leq \|u\|^2 \cdot \|v\|^2$$